USN | | | | | | | | | | [23TCSE208]

SECOND SEMESTER B.Tech EXAMINATION, MARCH-2024
INTRODUCTION TO PYTHON PROGRAMMING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

PART-A **5 X 2=10**

1.	**a.**	Explain with an example conditional expression in python.	L2	CO1	**02**
	b.	Differentiate lists and dictionaries in python with an example.	L2	CO4	**02**
	c.	Define modules in python.	L2	CO3	**02**
	d.	Explain data structures in pandas.	L3	CO2	**02**
	e.	Explain briefly removing duplicate data from DataFrame.	L1	CO5	**02**
		PART-B		**6 X 5=30**	
2.	**a.**	Differentiate between '/', '//', and '%' operators in Python?	L2	CO1	**05**
		OR			
	b.	Write a Python program to take a natural number as an input and find out whether the number is a prime or not.	L3	CO1	**05**
3.	**a.**	Differentiate between in-built functions and user-defined functions in Python using suitable examples.	L2	CO2	**05**
		OR			
	b.	Explain list comprehension with a suitable example.	L2	CO2	**05**
4.	**a.**	Discuss the difference between read(), readline(), and readlines() in Python file handling.	L3	CO3	**05**
		OR			
	b.	Describe the process to create and use your own module in Python.	L3	CO3	**05**
5.	**a.**	Write a program in Python to generate ten random integers. Comment your code for better understanding.	L3	CO2	**05**
		OR			
	b.	Explain how to perform interpolation using scipy module.	L3	CO3	**05**
6.	**a.**	Describe plotting using pandas with a suitable example.	L3	CO5	**05**
		OR			
	b.	Explain hierarchical indexing with an example.	L3	CO5	**05**
7.	**a.**	Explain recursion with a suitable example.	L3	CO2	**05**

		OR			
	b.	Differentiate between a Pandas series and DataFrame.	L3	CO5	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Explain various arithmetic operators in Python.	L2	CO1	**10**
		OR			
	b.	Discuss the comparison of two strings using any two comparison operators with an example.	L2	CO1	**10**
9.	**a.**	Differentiate between list, tuple, and dictionary using suitable examples.	L2	CO2	**10**
		OR			
	b.	Write a Python program to define functions for various logical operations (AND, OR, NOT, XOR) and call the defined functions to perform respective logical operation as per the user's choice taken as input.	L3	CO2	**10**
10.	**a.**	Describe the creation of package in Python.	L2	CO3	**10**
		OR			
	b.	Discuss various modes of file handling in Python.	L2	CO3	**10**
11.	**a.**	Write a function in python to count and display the total number of words in a text file.	L3	C04	**10**
		OR			
	b.	Write a Python program to create a pandas DataFrame (Students' Record) with 10 rows and 5 columns (Student_ID, Name, Age, Country, and Marks). After that, add a new column (Percentage) which will calculate the percentage of marks given in the Marks column.	L3	CO4	**10**
12.	**a.**	Discuss various ways of handling missing values in a dataset with an example.	L3	CO5	**10**
		OR			
	b.	Describe the concept of data wrangling in detail.	L3	CO5	**10**
13.	**a.**	Explain in detail various methods of creating Pandas DataFrame using lists of tuples.	L3	CO6	**10**
		OR			
	b.	Discuss plotting with seaborn in detail.	L3	CO6	**10**